

OXFORD

macroeconomics

INSTITUTIONS, INSTABILITY, AND INEQUALITY

WENDY CARLIN • DAVID SOSKICE

Macroeconomics for International Commerce

PART 1. Flipped Classroom – Week 4 Pre-Class Video Slides

Chapter 4: Expectations and Monetary Policy

How to use this video

Before class:

- Watch once straight through
- Try to catch the intuition, stop and think
- Bring one question to class
- In class: we will talk about more technical side and applications

Purpose of this video: *learn the intuition only. Technical derivations and diagrams will be done in class.*

Why expectations matter

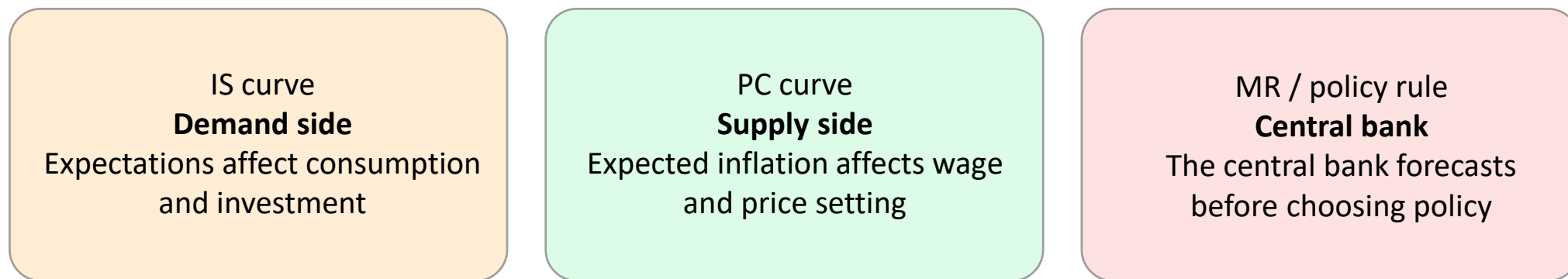
- Households form expectations about future income, prices and interest rates
- Firms form expectations about future demand, costs and profits
- Workers form expectations about inflation when bargaining over wages
- Central banks forecast inflation and output before setting policy

Expectations matter because many economic decisions are made before the future is known.



Expectations inside the 3-equation model

- The 3-equation model combines demand, supply, and policy.



Week 4 adds expectations to the framework you already saw in Week 3.

Inflation expectations in the Phillips Curve

$$\pi_t = \pi_t^E + \alpha(y_t - y_e)$$

- π_t^E is expected inflation
- The output gap affects wage pressure and price-setting
- If people expect higher inflation, wage demands and price increases may become higher too

This is why inflation can become persistent even after the original shock disappears.

Four ways to model expectations

Static expectations

Inflation is expected to stay constant

Adaptive expectations

Expected inflation follows past inflation

Rational expectations

Agents use all available information

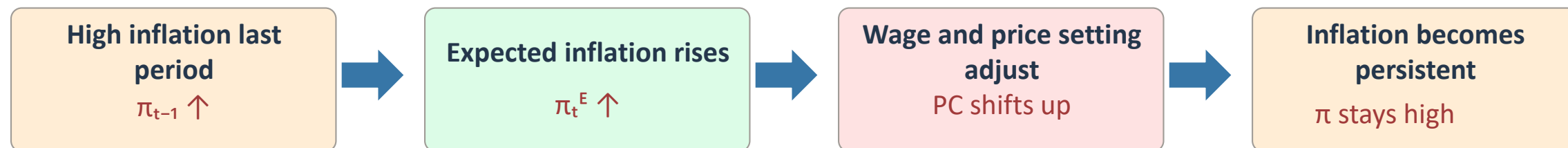
Anchored expectations

Expectations are tied partly to the central bank target

For this course, the most important contrast is adaptive vs anchored expectations.

Why adaptive expectations can be dangerous

- If inflation was high last period, people expect high inflation again.



Adaptive expectations can turn a temporary inflation shock into repeated inflation pressure.

Why anchored expectations help policy

- Anchored expectations mean people still believe the central bank will return inflation to target.

$$\pi_t^E = \chi\pi^T + (1-\chi)\pi_{t-1}$$

χ = credibility of the central bank

If $\chi = 1$, suggesting that the CB is fully credible, then the PC is anchored to π^T

A special case of static expectations PC with $\bar{\pi} = \pi^T$

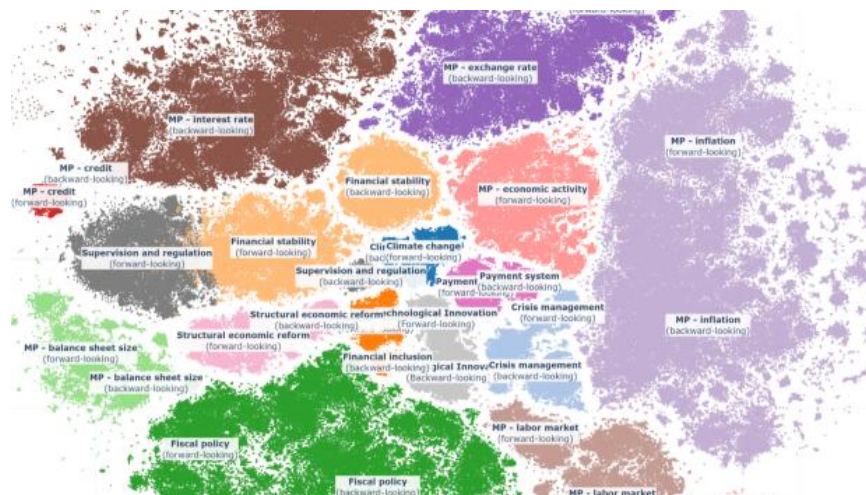
If $\chi = 0$, then $\pi_t^E = \pi_{t-1}$ just like adaptive expectations

- Temporary inflation shock
- Expectations stay near target
- Smaller wage-price response
- Lower output cost of disinflation

The more credible the central bank is, the more expectations stay close to the inflation target.

Credibility and central bank communication

- Inflation targeting works better when people believe the target
- Transparency helps households, firms and markets understand policy decisions
- Independence protects monetary policy from short-run political pressure
- Communication is part of monetary policy, not only explanation after the fact



A visual representation of central bank communication over a century of data from recent staff working paper, "From Text to Quantified Insights"

"How AI is Reshaping Central Bank Communication Analysis,"
IMF Market Insights

For firms engaged in international commerce, credibility affects inflation expectations, interest rates and exchange-rate risk.

Policy lesson: expectations can change outcomes

- The same policy can have different effects depending on what people expect.

Weak credibility

Inflation expectations rise

PC shifts up more easily

Policy becomes more costly

Strong credibility

Expectations stay anchored

PC shifts less

Policy becomes less costly

This is the intuition behind the Lucas critique: people change behavior when policy regimes change.

Readiness check

After watching, you should be able to answer these before class.

1. Why do expectations matter in macroeconomics?
2. In the Phillips Curve, expected inflation matters because...?
3. What are adaptive expectations?
4. What are anchored inflation expectations?
5. Suppose inflation rises from 2% to 5% because of an energy-price shock. Under adaptive expectations, what may happen next?
6. Why does central bank credibility matter?
7. Short answer: What is the difference between adaptive and anchored expectations?
8. Short answer: Why can anchored expectations reduce the output cost of fighting inflation?

For class, remember the causal chain: Inflation shock or policy change → people form expectations about future inflation → expectations affect wage-setting and price-setting → Phillips Curve shifts or stays anchored → inflation becomes persistent or returns to target → central bank response becomes more or less costly.

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Module code

PART 2. Full In-Class Slides

Chapter 4: Expectations
and Monetary Policy

Objectives:

By the end of this chapter, students should understand the following:

- Why economic agents form expectations
- Four different ways of modelling inflation expectations
- The role of expectations in the 3-equation model
- The Lucas Critique and macroeconomic policy
- Inflation bias
- Anchored inflation expectations and implications of hysteresis

Introduction: expectations

Expectations are formed by households, firms and the policy maker. Expectations influence consumption, investment, wage-setting and policy decisions

This is reflected in the 3-equation model:

IS Curve

- Tobin's Q: Firms form expectations of future profits
- Permanent Income Hypothesis: Households form expectations over their future income

PC Curve

- Wage setters form inflation expectations (π_t^E)

MR Curve

- Policy maker forecasts inflation next period

Introduction: risk and uncertainty

Risk: Known probabilities can be attached to future outcomes

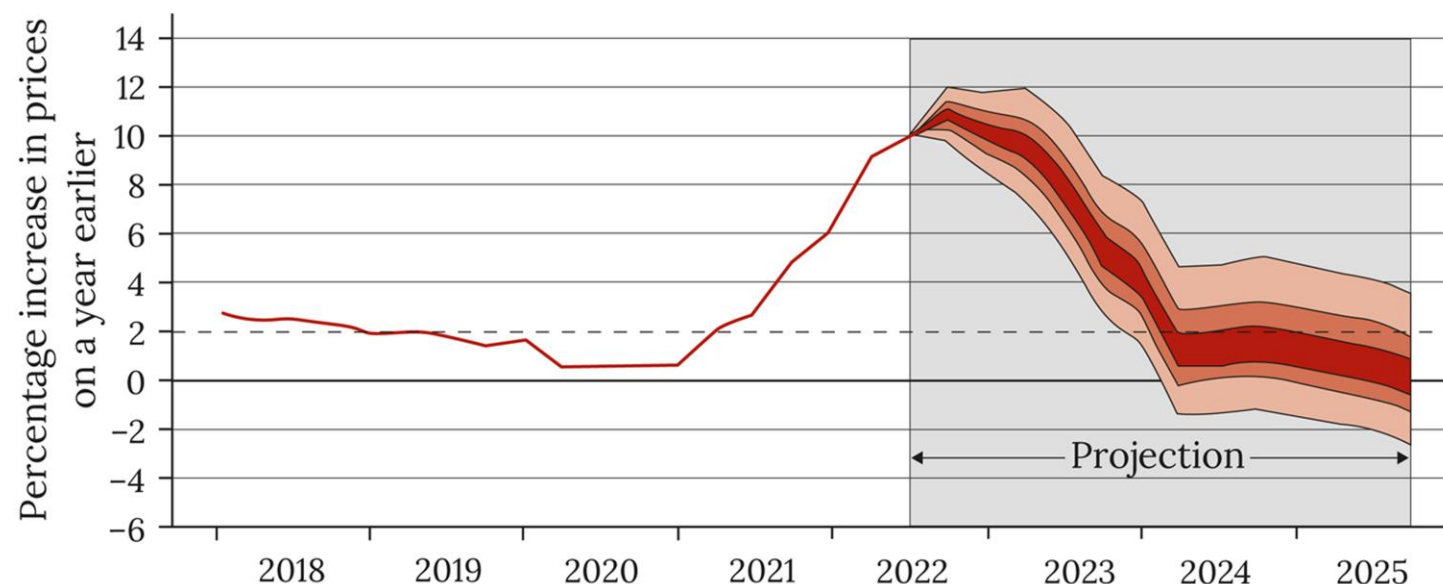
Uncertainty: Impossible to assign probabilities to known outcomes; there might be unknown outcomes

- Agents form expectations differently when they face risk compared to uncertainty
- E.g., BoE forecasts inflation by attaching subjective probabilities to different scenarios:

Figure 4.1 Bank of England's CPI inflation projection.

Note: If economic circumstances identical to today's were to prevail on 100 occasions, the inflation rate is expected to lie within the darkest central band on only 30 of those occasions. Outturns are expected to lie within each pair of the lighter areas on 30 occasions and to lie somewhere within the fan on 90 occasions.

Source: Chart 1.4 (page 22), *Bank of England Monetary Policy Report*, November 2022.



Introduction: inflation bias

If the central bank targets an output level above equilibrium, then there is inflation bias

- The results about inflation bias are important in understanding why central bank independence and inflation targeting were adopted in the 1990s
- Two modifications of the model mitigate inflation bias
 - Anchored inflation expectations: inflation expectations are a weighted average of past inflation and the central bank's inflation target
 - Hysteresis: allow equilibrium output to be affected by past output levels
- A model with anchored expectations *and* hysteresis allows the CB to lead the economy to a new equilibrium with higher output and inflation at target

Phillips Curve (PC), expectations and inflation: settings

We are concerned with agents' expected inflation in period t given information set $t - 1$:

$$E_t(\pi_t | \Omega_{t-1}) \equiv \pi_t^E$$

- Ω_{t-1} contains the outcome of all the macroeconomic variables up to and including period $t - 1$

We use a standard PC that allows us to discuss different ways of modelling inflation expectations:

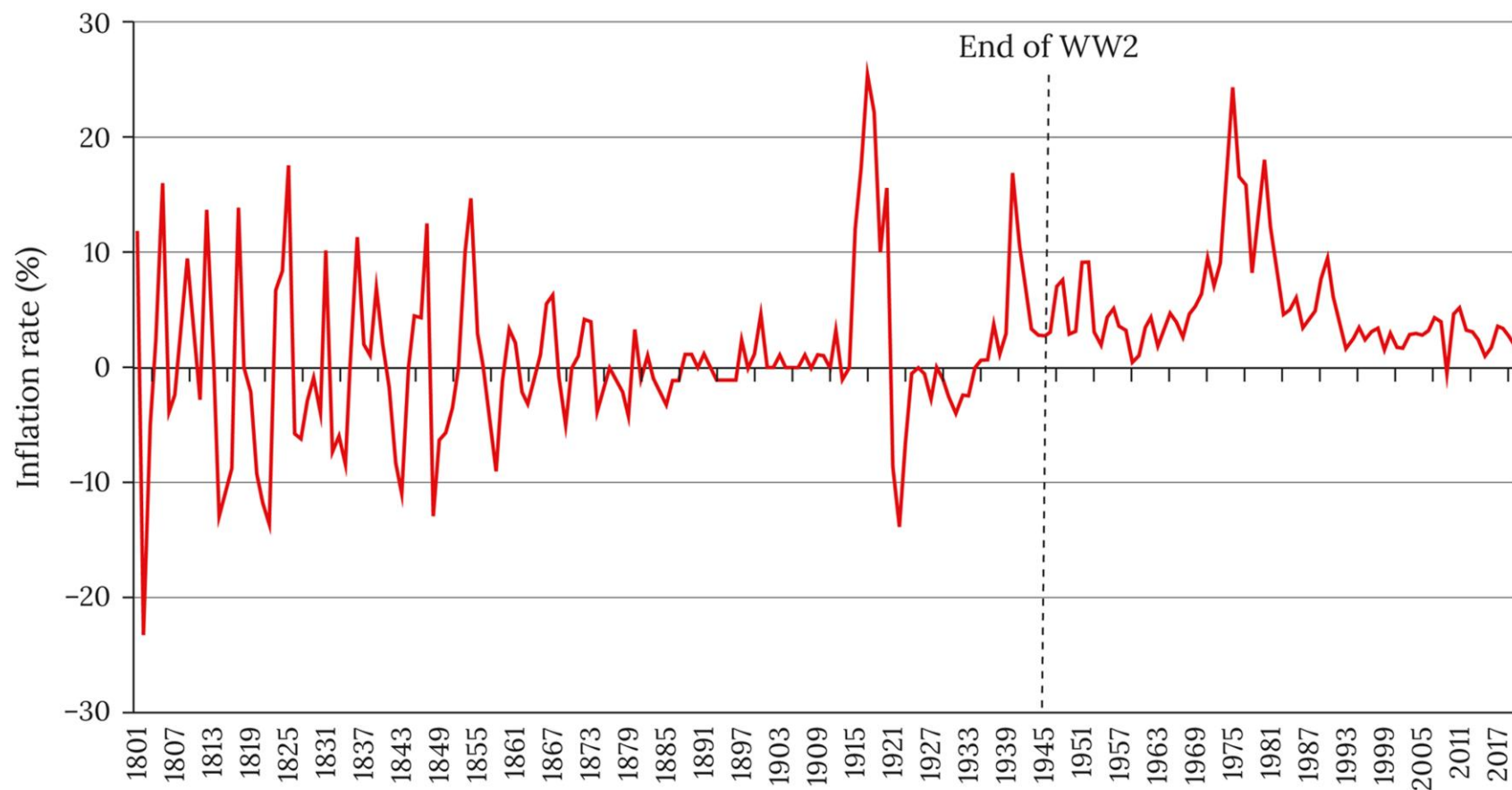
$$\pi_t = \pi_t^E + \alpha(y_t - y_e)$$

PC, expectations and inflation: types of inflation expectation

Ways of modelling inflation expectations:

- Static expectations: agents expect inflation to stay constant
- Adaptive expectations: agents expect inflation to be the same as the previous period
- Rational expectations: agents make use of all information available
- Anchored expectations: agents expect inflation to be a weighted average of the CB's inflation target and lagged inflation

PC, expectations and inflation: post-war UK inflation



UK inflation in the post-war period:

- 1950s-60s: Positive but stable inflation
- 1960s-70s: high inflation

Figure 4.2 UK inflation: 1801–2021.

Note: Variable shown is the annual percentage change of the long-term indicator of prices of consumer goods and services.

Source: UK Office for National Statistics (July 2022 release).

PC, expectations and inflation: post-war UK inflation

Positive but stable inflation in 1950s and 1960s:

- The governments were pursuing activist demand management policies. They feared unemployment and deflation more than inflation

This is consistent with static expectations:

$$\begin{aligned}\pi_t &= \pi_t^E + \alpha(y_t - y_e) \\ &= \bar{\pi} + \alpha(y_t - y_e)\end{aligned}\quad \text{(static expectations Phillips curve)}$$

- As $\bar{\pi}$ is a constant, inflation only fluctuates with the business cycle
 - Inflation falls in recession and inflation increases in expansion
- The static expectations Phillips curve only shifts if there is a supply-side shock that changes y_e

PC, expectations and inflation: post-war UK inflation

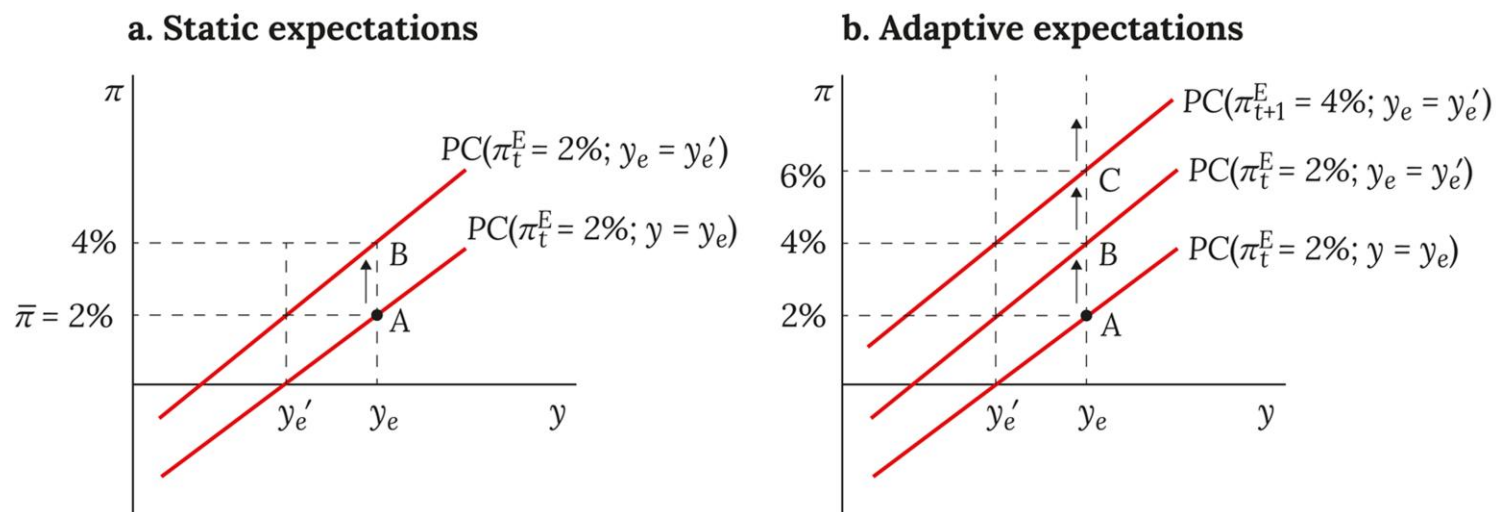
Rising inflation from the end of 1960s to 1970s:

- Negative supply shocks led to rise in equilibrium unemployment

At $y = y_e > y'_e$:

- Static expectations can explain higher inflation but not periods of rising inflation
- Adaptive expectations: Wage-setters update their expectations each period to correct for the erosion of expected real wages → PC shifts upwards each period

Figure 4.3 Static and adaptive expectations: the government attempts to maintain low unemployment after a negative supply-side shock.



PC, expectations and inflation: output-inflation trade-off

$$\pi_t = \bar{\pi} + \alpha(y_t - y_e) \quad (\text{static expectations Phillips curve})$$

- Hence: $\pi_t - \bar{\pi} = \alpha(y_t - y_e)$
- When there's a positive output gap, inflation is higher but not rising
- Output-inflation trade-off: policy makers can operate at $y > y_e$ if they allow a higher level of inflation

$$\pi_t = \pi_{t-1} + \alpha(y_t - y_e) \quad (\text{adaptive expectations Phillips curve})$$

- Hence: $\pi_t - \pi_{t-1} = \Delta\pi_t = \alpha(y_t - y_e)$
- There is always upward pressure on inflation if output stays above equilibrium
- No long-run output-inflation trade-off under adaptive expectations: policy makers cannot choose a higher level of constant inflation to generate $y > y_e$

PC, expectations and inflation: output-inflation trade-off

Absence of a long-run output-inflation tradeoff is supported by the experience of high inflation and high unemployment in many developed economies

- the stagflation of the 1970s

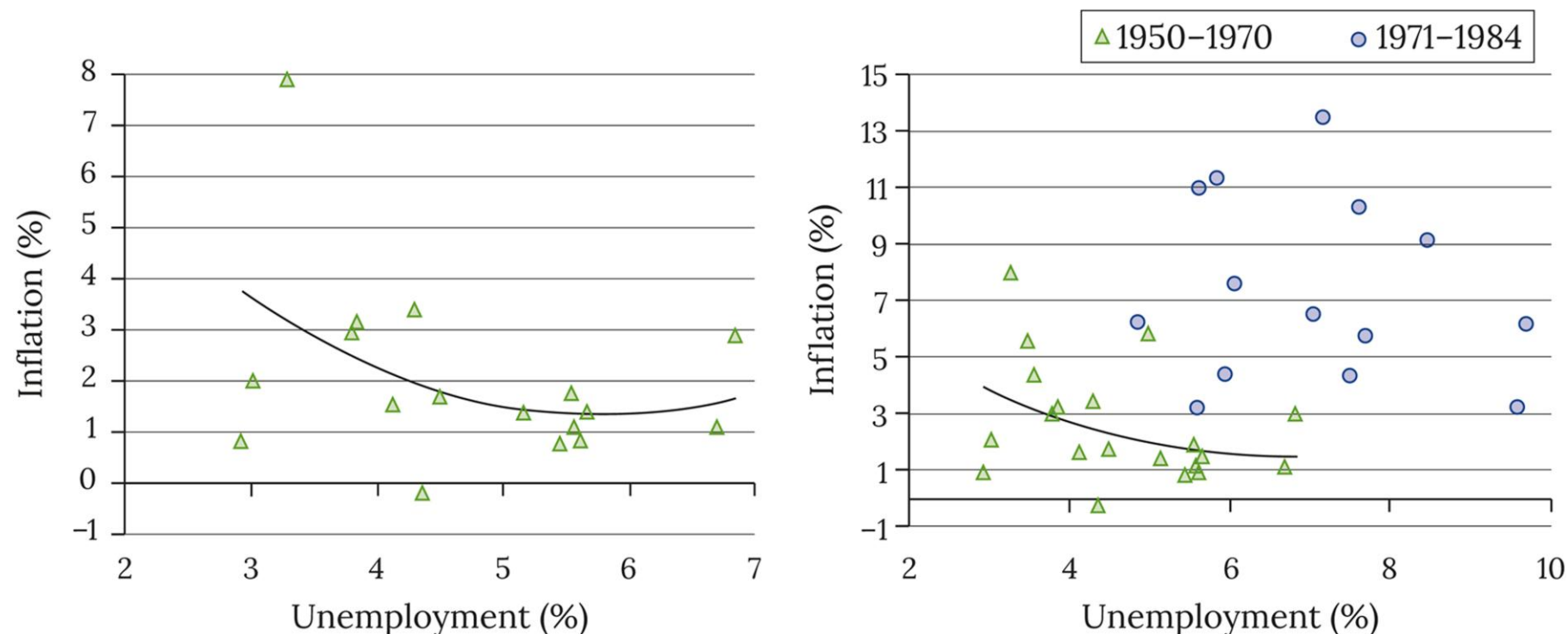


Figure 4.4 US empirical Phillips curves: (left) 1950-70, (right) 1950-70 and 1971-84. Note: The black curves on the graphs are quadratic regression lines. Source: US Bureau of Labor Statistics (accessed July 2020).

PC, expectations and inflation: rational expectations

Agents with rational expectations in the 3-equation model:

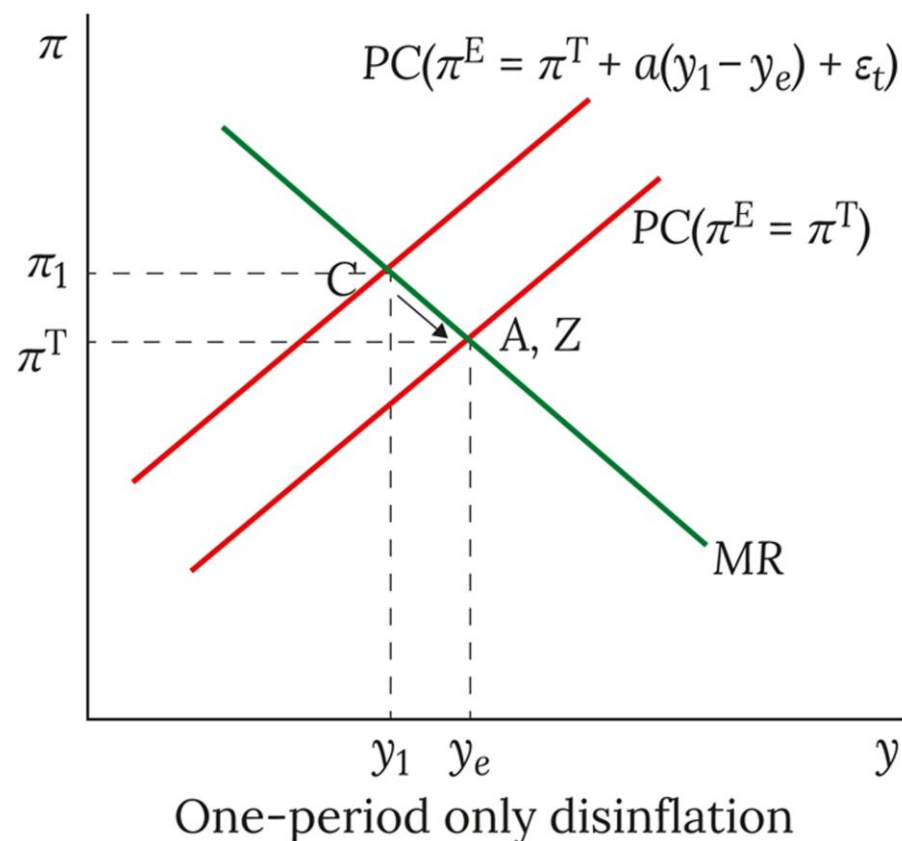
- They have full knowledge of the model (like the central bank)
- W- and P-setters are forward-looking and fully informed but W, P not fully flexible
- Wages set periodically

$$\pi_t = \pi^T + \alpha(y_t - y_e) + \epsilon_t \quad (\text{rational expectations Phillips curve})$$

- ϵ_t is a random shock to inflation with a mean of zero
- Only unanticipated shocks (ϵ_t) to inflation and the output gap cause inflation to be different from its expected value

PC, expectations and inflation: rational expectations

a. Rational expectations in the 3-equation model

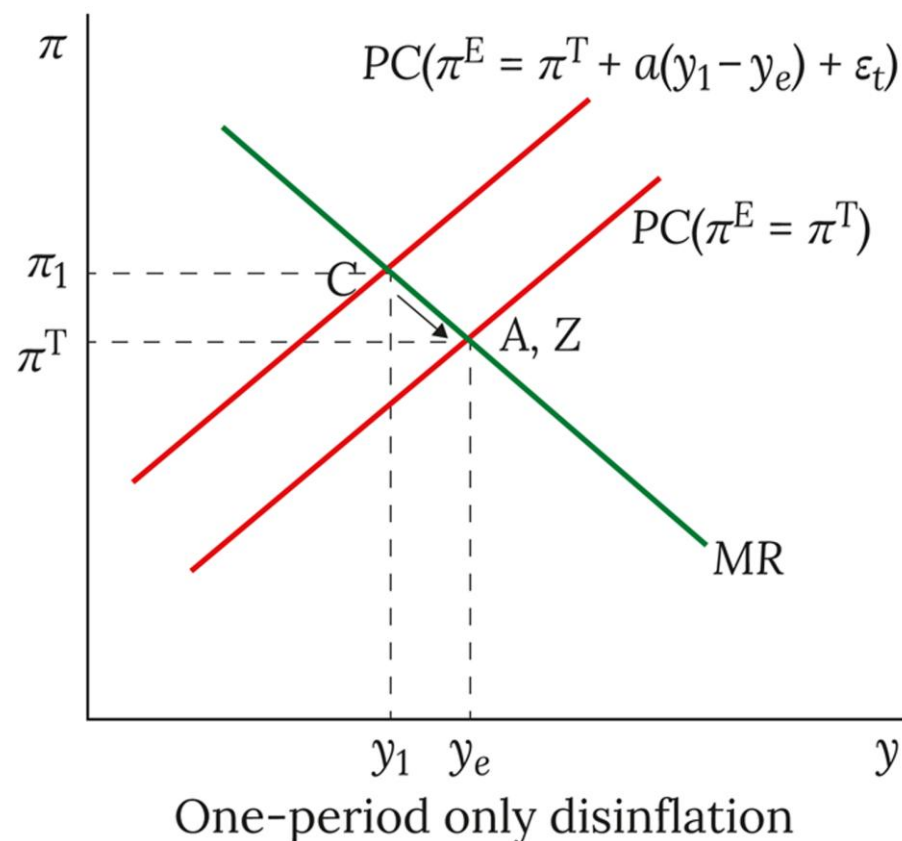


- A change in product market conditions enables firms to increase their markup
- The shock and the model are common knowledge
- Wage setting has already occurred so wage setters cannot respond to the news
- Price setters and CB know the MR but need to work out the new PC to choose π_1 and y_1 simultaneously.

Figure 4.5(a) Adjustment to an inflation (markup) shock in the 3-equation model under rational and adaptive expectations. Visit www.oup.com/he/carlin-soskice to explore rational and adaptive expectations with Animated Analytical Diagram 4.5.

PC, expectations and inflation: rational expectations

a. Rational expectations in the 3-equation model



- The new PC:

The key equations in the new PC with markup shock are:

$$(\Delta W / W)_1 = (\Delta W / W)_0 = \pi^T \quad (\text{wage inflation})$$

$$\frac{\lambda(1 - \mu_0) - \lambda(1 - \mu_1)}{\lambda(1 - \mu_0)} = \frac{\mu_1 - \mu_0}{1 - \mu_0} \equiv \varepsilon_1 \quad (\text{markup inflation shock})$$

New rational expectations PC:

$$\pi_1 = \pi^T + \alpha(y_1 - y_e) + \frac{\mu_1 - \mu_0}{1 - \mu_0}$$

$$\pi_1 = \pi^T + \alpha(y_1 - y_e) + \varepsilon_1 \quad (\text{rational expectations PC; 3-equation model})$$

- Given the new PC and the MR, the economy moves to point C.

Figure 4.5(a) Adjustment to an inflation (markup) shock in the 3-equation model under rational and adaptive expectations. Visit www.oup.com/he/carlin-soskice to explore rational and adaptive expectations with Animated Analytical Diagram 4.5.

PC, expectations and inflation: rational vs adaptive expectations

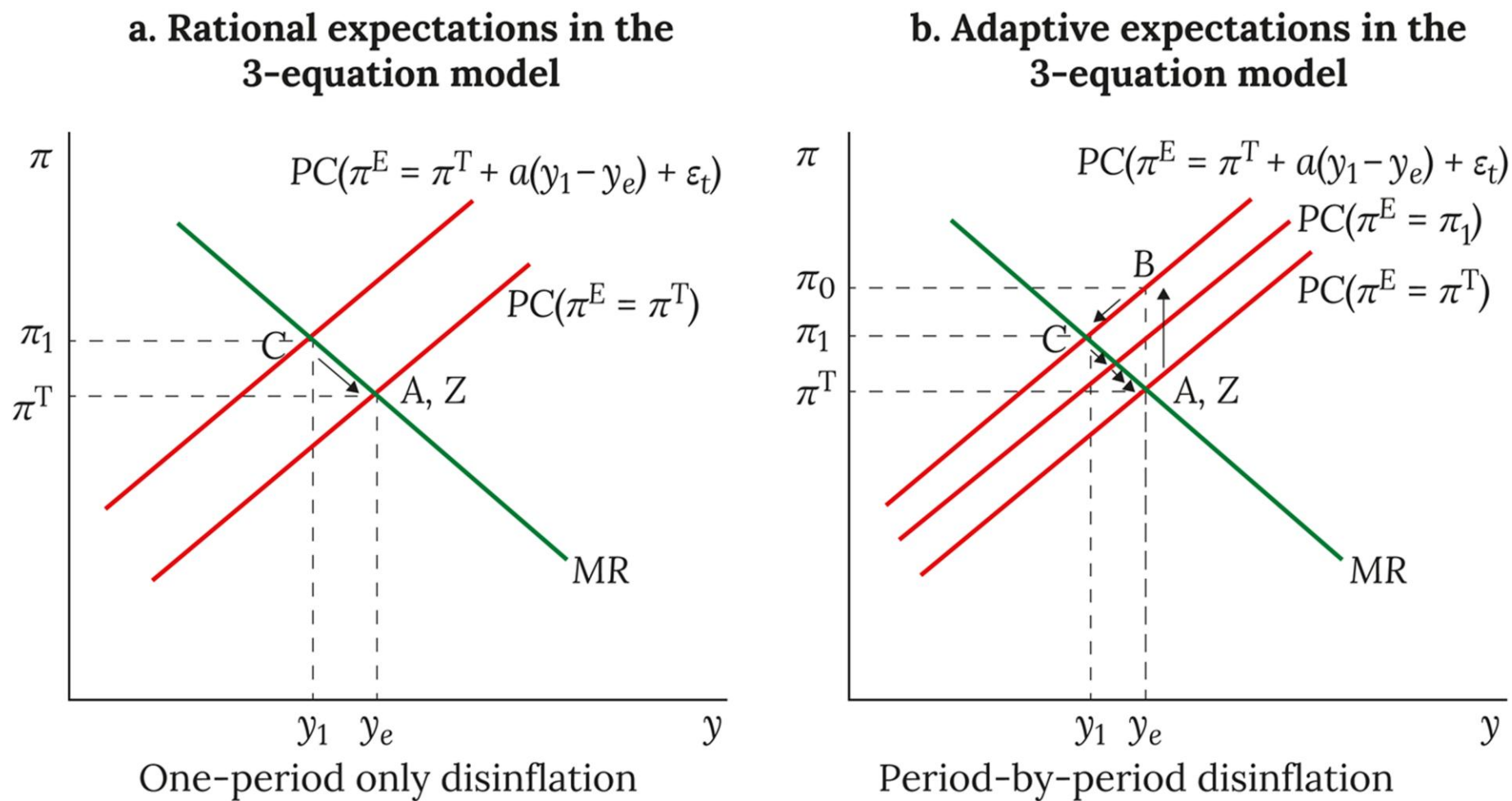


Figure 4.5 Adjustment to an inflation (markup) shock in the 3-equation model under rational and adaptive expectations. Visit www.oup.com/he/carlin-soskice to explore rational and adaptive expectations with Animated Analytical Diagram 4.5.

PC, expectations and inflation: implications of rational expectations

- Expected and actual inflation are the same

Implication: Costless shift of inflation target

- If the central bank wants to reduce its inflation target, it can announce the new target before the wage-setting round. Both MR_{REH} and PC shift down and the change is costless
- Under adaptive expectations, costly period-by-period adjustment is required to attain lower inflation target

Simpler job for the CB when *both* expectations and wage and price setting change

- If the assumption of perfectly flexible wages and prices is added, then the CB only needs to set the inflation target without having to implement active monetary policy (no MR curve)

PC, expectations and inflation: rational vs adaptive expectations

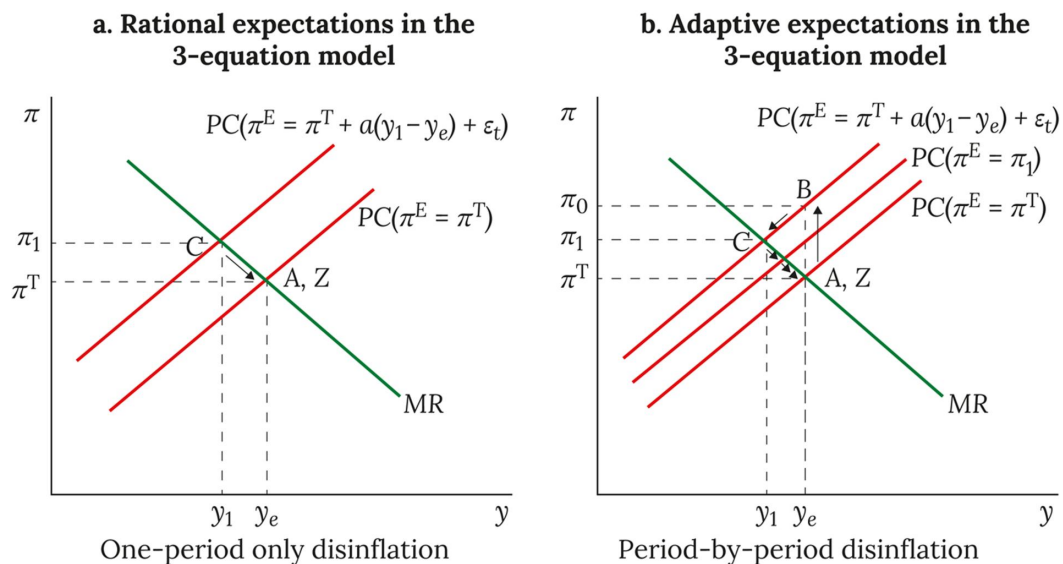


Figure 4.5 Adjustment to an inflation (markup) shock in the 3-equation model under rational and adaptive expectations. Visit www.oup.com/he/carlin-soskice to explore rational and adaptive expectations with Animated Analytical Diagram 4.5.

- Under REH, decisions of price setters and CB are simultaneous: $A \rightarrow C$
- Under AEH, decisions are sequential.
- Under AEH, given that wages have already been set, price setters raise prices according to the new PC and the economy moves from A to B.
- Given the new PC, the CB chooses point C on the MR.
- The markup shock disappears before the next round of wage setting, and wages are set using $PC(\pi^E = \pi_1)$; the CB chooses point D. The economy moves period-by-period back to equilibrium at Z.

The Lucas critique

The Lucas critique:

- Models that rely on relationships found in historical data are problematic
- Relationships in historical data are conditional on past policy regimes, and will break down if the policy regime changes
 - Economic agents change their behaviour in response to the new policy regime
- An example in the 3-equation model under REH:
 - Government spending effects and revoking CB independence

The Lucas critique: government spending effects - example

The government wants to boost its popularity before the election:

- It increases spending after the CB has set the interest rate for the period
- Hence, $y_t > y_e$ and yet the CB cannot react until the next period
- Under REH, wage and price setters assume $\pi_t^E = \pi^T$ and believe that $y_t = y_e$, therefore inflation stays on target
- In the next period, the CB reacts by increasing interest rate
- Inflation is now at target and output falls back to equilibrium
- If wage and price setters have rational expectations, the best a fiscal stimulus can do is a 1 period increase in y
- As a result, the data shows a one period rise in output without a rise in inflation above target

The Lucas critique: revocation of CB's independence - example

The new government observes this past data:

- The previous data suggests that government spending boosts y without increasing π
- Thinking that the CB is (unnecessarily) conservative, the new government takes away CB's independence and prohibits it from raising interest rate when government spending increases
- Since the CB cannot react, increasing government spending causes inflation to increase next period
- Inflation increases without limit

Hence, this example illustrates that forecasting policy outcomes using historical relationships can produce poor economic outcomes

Inflation bias

The policy makers may want to target an output that is above equilibrium:

- Their incentives: political concerns for the next election; desire to increase social welfare by reducing involuntary unemployment
- Hence, their loss function now involves the output target y^T :

$$L = (y_t - y^T)^2 + \beta(\pi_t - \pi^T)^2$$

Note that $y^T > y_e$

- Despite the good intention, this gives rise to inflation bias: output stays at equilibrium, but inflation is above target
- This outcome is unambiguously worse than if a CB simply targets π^T and a zero output gap

Inflation bias in the 3-equation model

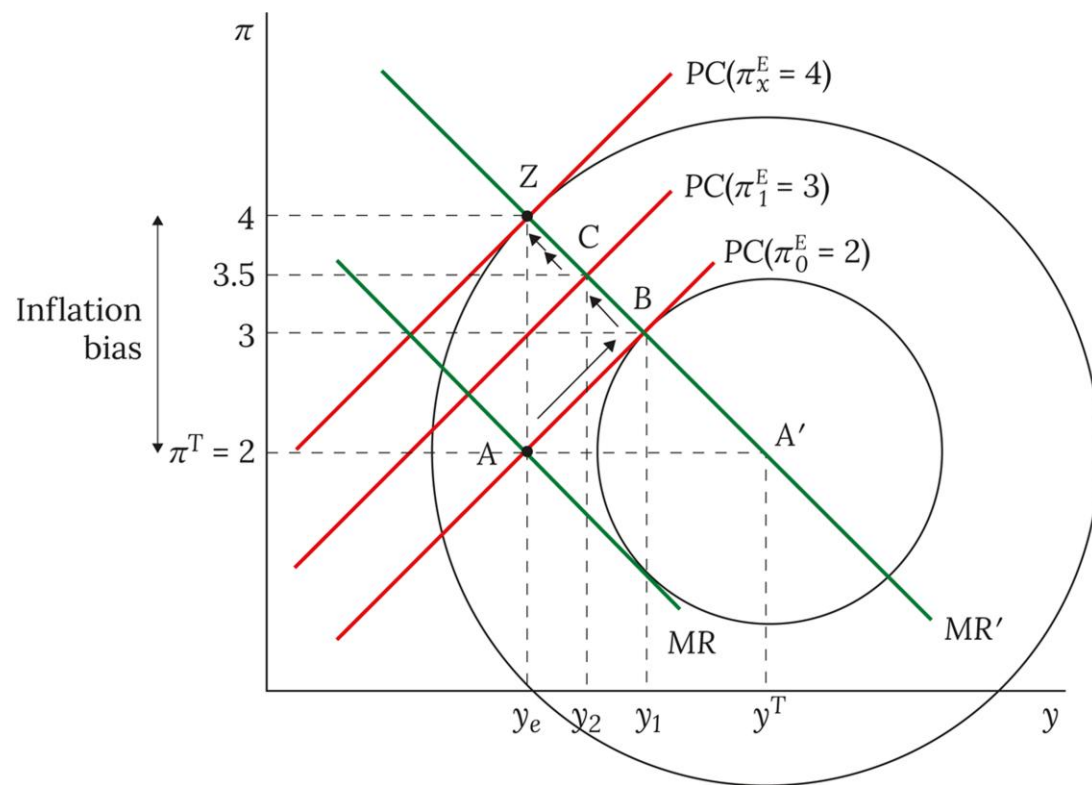


Figure 4.6 The inflation bias. Visit www.oup.com/he/carlin-soskice to explore inflation bias with Animated Analytical Diagram 4.6.

Assume adaptive expectations:

- MR shifts right to MR' and goes through the CB's new bliss point A' with π^T and y^T
- Period 0: CB forecasts that PC will not move the next period. Using MR' , the CB chooses B
- Period 1: there is inflationary pressure as the output is above equilibrium. PC shifts up and the CB chooses C
- Period 2 onwards: again, there is inflationary pressure and PC shifts up. CB reoptimizes by raising the interest rate. The adjustment process stops until the economy reaches Z with y_e and inflation above target

Inflation bias: under adaptive expectations

We can derive the inflation bias:

- With the new loss function and PC, we have the new MR:

$$(y_t - y^T) = -\alpha\beta(\pi_t - \pi^T)$$

- At equilibrium, $y_t = y_e$ and $\pi_t = \pi_{t-1}$. Hence:

$$y_e = y^T - \alpha\beta(\pi_{t-1} - \pi^T)$$

$$\Rightarrow \pi_t = \pi^T + \underbrace{\frac{(y^T - y_e)}{\alpha\beta}}_{\text{inflation bias}}$$

(inflation bias under AEH)

- Inflation bias is positive whenever $y^T > y_e$

Size of the inflation bias is larger:

- The less inflation averse the CB is (steeper MR curve, low β)
- The lower the output sensitivity of inflation is in the PC (low α ; large fall in y needed for low π , so CB just aims for higher π)

Inflation bias: under rational expectations

Inflation bias also arises under rational expectations:

- y^T announced: instantaneous adjustment from point A to Z in slide 22
- Forward-looking, rational agents know that given the new MR curve, the medium-run equilibrium is at Z
- This generates the same inflation bias as under adaptive expectations

The CB would have been better off committing to A , but they cannot do so due to time inconsistency and credibility issue

- Rational agents hold inflation expectation $\pi^E = \pi_Z$ which is consistent with the CB's loss function
- If those agents hold $\pi^E = \pi_A$, then the CB has an incentive to choose $y > y_e$
- Hence, a CB with a preference for $y^T > y_e$ cannot credibly commit to y_e

Inflation bias: mitigating inflation bias

Delegation:

- Delegating monetary policy to a CB which has no incentive to set output above equilibrium
- Longer term tenures for policy makers ensure that they take into account policy impact over a longer period of time
- Empirical evidence suggests that central bank independence reduces inflation

Reputation Building (analysis with Game Theory):

- 2 periods: The CB builds a 'tough' reputation on inflation (i.e., sets y^T closer to y_e) to keep inflation expectations low $\rightarrow$ bigger gain from increasing y in the second period
- Many periods: The benefits from behaving 'tough' increase
 - Period 1's situation is repeated until the last period, since there are high gains from keeping π^E low

Anchored inflation expectations

Anchoring inflation expectations:

- Central bank communication is used to keep inflation expectations (π^E) anchored at the inflation target (π^T)
- Under anchored expectations, expected inflation is a weighted average of the inflation target and lagged inflation:

$$\pi_t = [\chi\pi^T + (1 - \chi)\pi_{t-1}] + \alpha(y_t - y_e) \quad (\text{anchored expectations PC})$$

where $\pi_t^E = \chi\pi^T + (1 - \chi)\pi_{t-1}$ and χ represents the credibility of CB

- If $\chi = 1$, suggesting that the CB is fully credible, then the PC is anchored to π^T
 - A special case of static expectations PC with $\bar{\pi} = \pi^T$
- If $\chi = 0$, then $\pi_t^E = \pi_{t-1}$ just like adaptive expectations

Anchored inflation expectations: under inflation shock

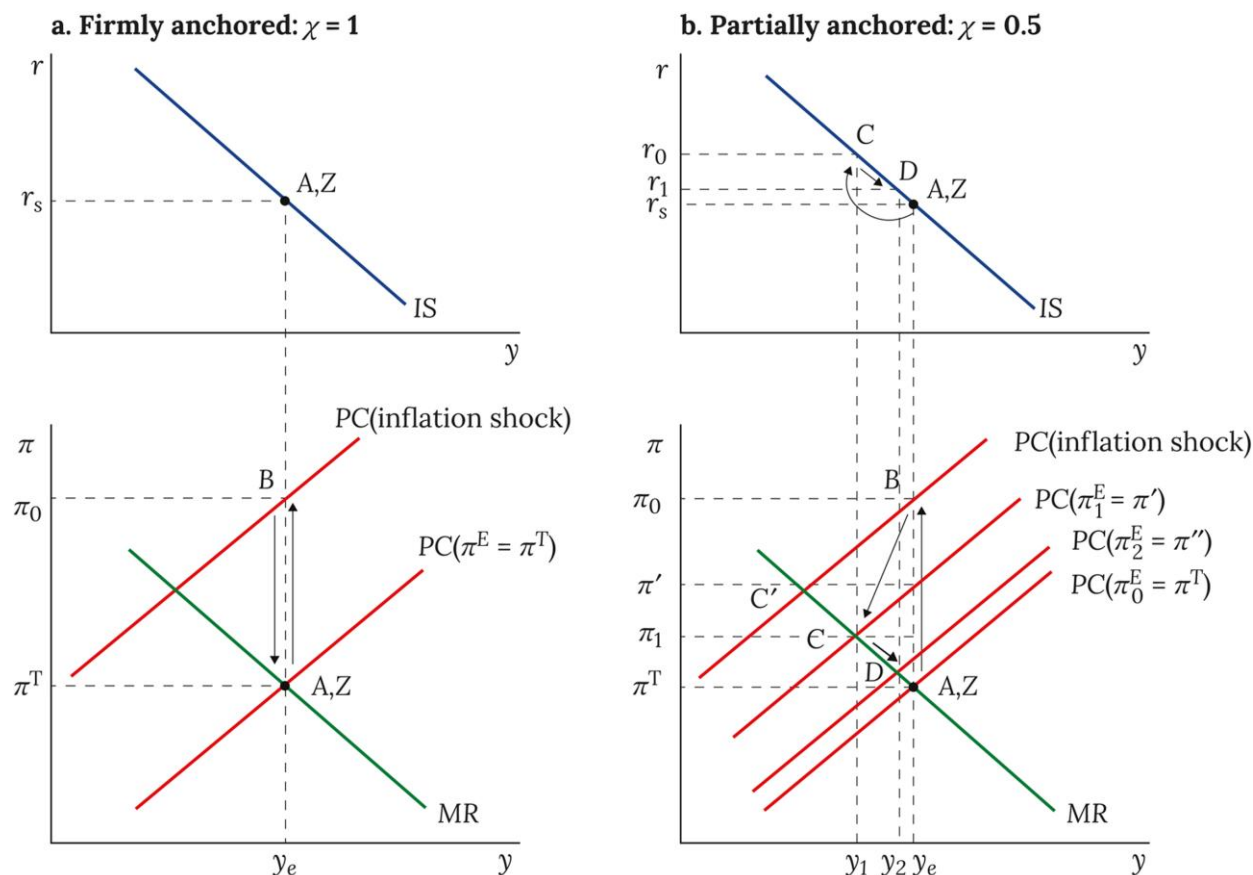


Figure 4.7 Varying the level of central bank credibility, using the example of an inflation shock:

a. Fully anchored: $\chi = 1$

b. Partially anchored: $\chi = 0.5$.

When $\chi = 1$: firmly anchored at π^T

- Only 1 period effect of high inflation $\rightarrow$ PC reverts to $PC(\pi^E = \pi^T) \rightarrow$ CB does not need to change $r \rightarrow$ disinflation is costless

When $\chi = 0.5$: partially anchored at π^T

- After the shock, CB needs to raise r , but to a lesser extent than $\chi = 0 \rightarrow$ fall in output is lower, adjustment to equilibrium is quicker
- PC shifts down to $PC(\pi_1^E = \pi')$ instead, economy moves to point C

Anchored inflation expectations: under inflation shock

Compared to what we have seen under adaptive expectations in Ch.3, any degree of anchoring of inflation expectations improves welfare:

- Output and inflation return to equilibrium more rapidly

Late 1990s: The developed world central banks have realized the importance of independence, credibility and transparency:

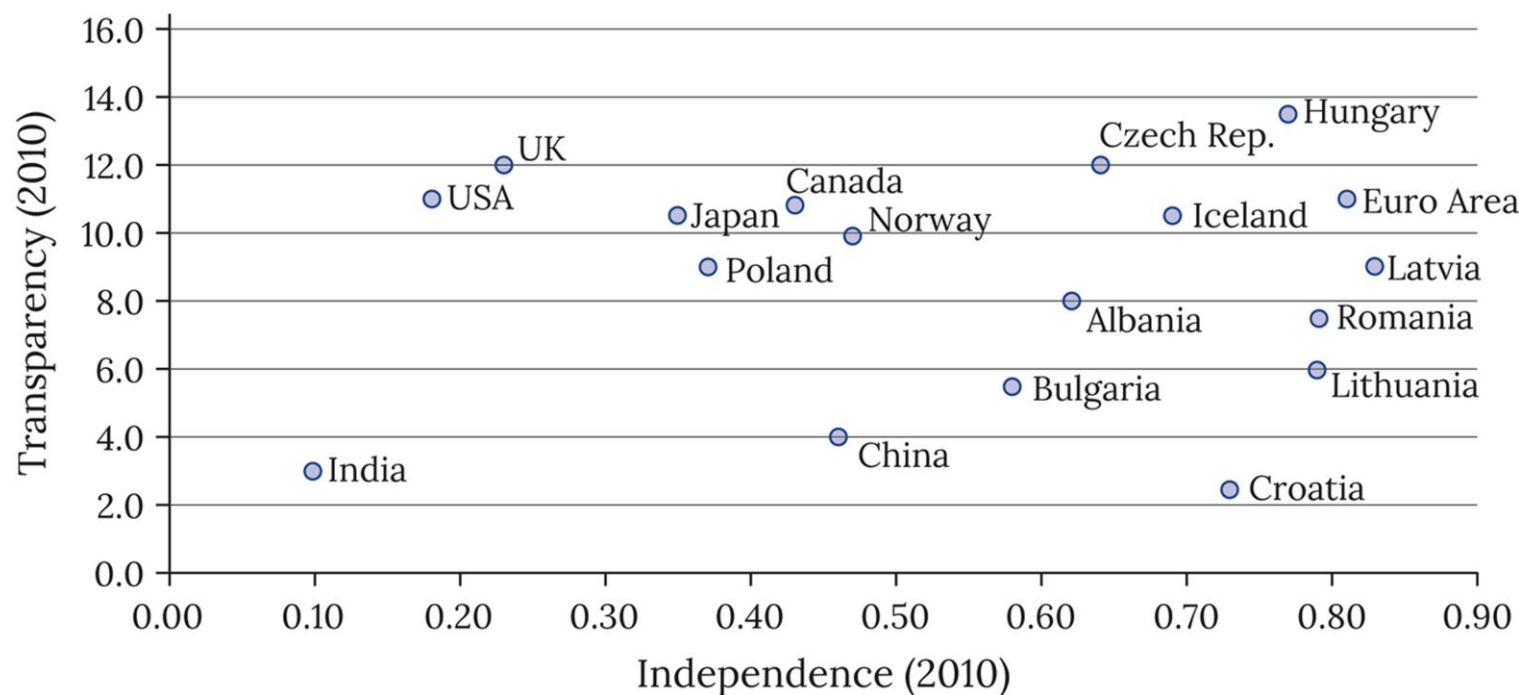


Figure 4.8 Central bank independence and transparency (in 2010).
Source: Dincer and Eichengreen (2014).

Hysteresis: 'past output level affects equilibrium output'

Although inflation bias suggests targeting a level of output above equilibrium is impossible, there is a state of the economy where an output target above equilibrium is both achievable and desirable:

- An economy that has experienced a deep recession experiences hysteresis, which means equilibrium output falls
- If a long recession reduces equilibrium output and if the damage can be reversed by higher economic activity, then a policy that targets the initial equilibrium level of output has merit

Hysteresis:

Experience during the early 2010s suggests that hysteresis indeed accompanies a prolonged downturn:

- Continuous drop in estimates of potential GDP (equilibrium output in the model)

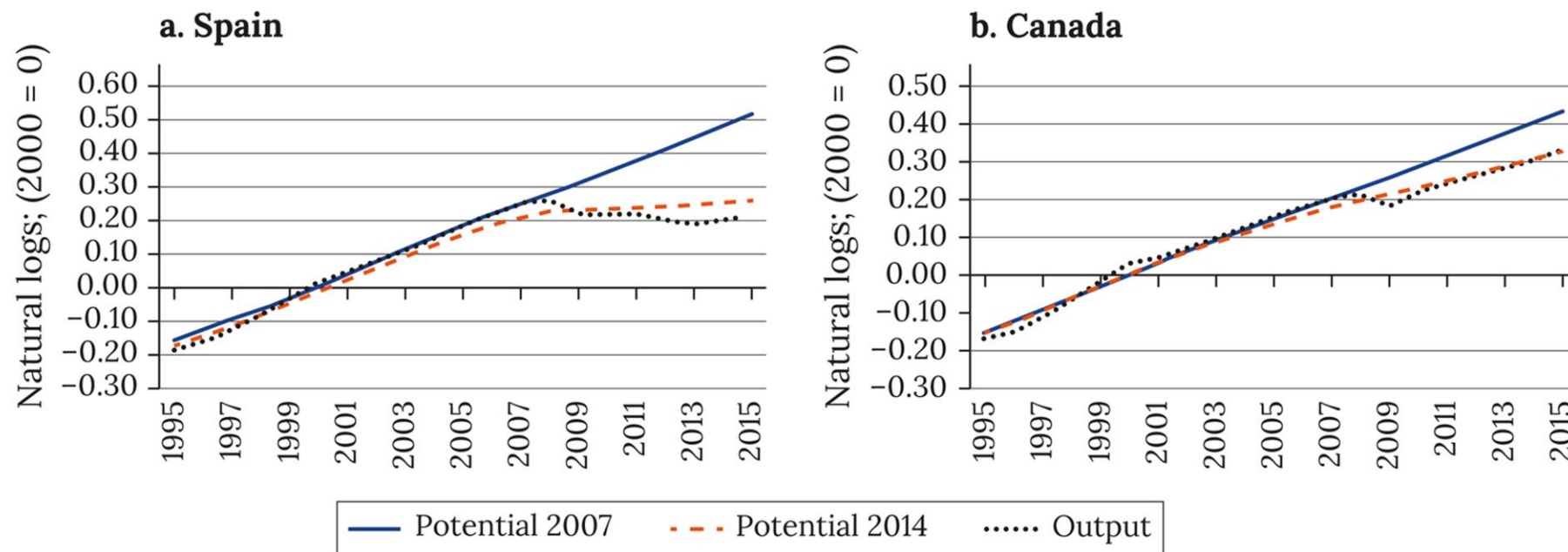


Figure 4.9 This replicates 2 panels of Figure 2 from Ball (2014). Data on potential output was used from the 2007 *OECD Economic Outlook*, and the 2015 *OECD Economic Outlook* was used to measure actual output. This uses log of potential output, normalizing the year 2000 to 0. The 2007 projections were extended beyond 2009 by using average growth of log potential output in previous years, as in Ball (2014).

Hysteresis: hysteresis mechanisms

Recession itself can shift WS curve upward:

- Workers are laid off during the recession → their skills erode (e.g., lose touch with new technologies and work practices) → they lose competitiveness in the labour market → they give up job search and labour participation rate decreases → upward shift in the WS curve → lower equilibrium output
- Empirical evidence for the mechanism:
 - Yagan (2019): the long-term scarring effects of the recession
 - Heathcote et al. (2020): the scarring effects of recessions increase inequality
- This mechanism helps explain the government's policy of subsidizing wages to keep workers attached to their employers during the Covid-19 crisis

Hysteresis: anchored inflation expectations (Case 1, without hysteresis)

With anchored expectations PC, there's a long-run inflation-output tradeoff that depends on χ (the weight on the inflation target):

- We can illustrate with a simulation:
 - The economy begins with $y_e = 98$ and $\pi^T = 2\%$; period 5: the CB adopts $y^T = 100$
- Inflation-output tradeoff: With positive values of χ , the economy stabilizes again at higher output and inflation above target. The larger the value of χ , the smaller the increase in medium-run inflation above the target

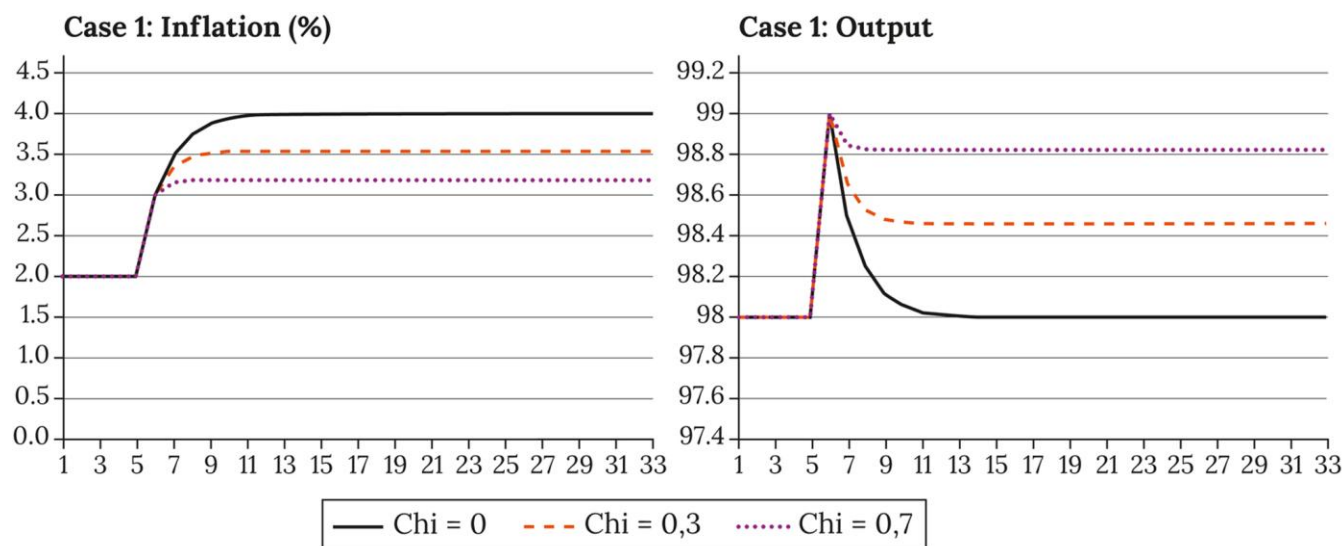


Figure 4.10 Case 1 (comparing the degree of expectation anchoring). The x-axis represents the number of periods elapsed, starting in period 1. The y axis on the left-hand side graphs represents inflation in the model, in percent increase in prices per period, while the y-axis on the right-hand side graphs represents output per period relative to initial output of 98.

Hysteresis: anchored inflation expectations (Case 1, without hysteresis)

However, in Case 1, the CB is exploiting the credibility of its inflation target:

- Over time, agents lose confidence in the central bank's inflation target and inflation bias would reappear
- When $\chi = 0$, output returns to 98 and inflation is above target in the new equilibrium

Hence, there is a paradox in Case 1

- On its own, anchored inflation expectations is not enough to generate an outcome where output is above equilibrium and inflation is at target
- We need hysteresis as well as anchoring

Hysteresis: anchored expectations and hysteresis (Case 2)

If the past output level affects equilibrium output, there is hysteresis and then the CB can achieve both its inflation and output target:

- Conduct the same simulation with inflation expectations anchored at π^T :
 - The economy converges to a new equilibrium with $\pi = 2\%$ and $y = 100$
 - Because of hysteresis, higher economic activity improves the supply side, which allows equilibrium output to rise

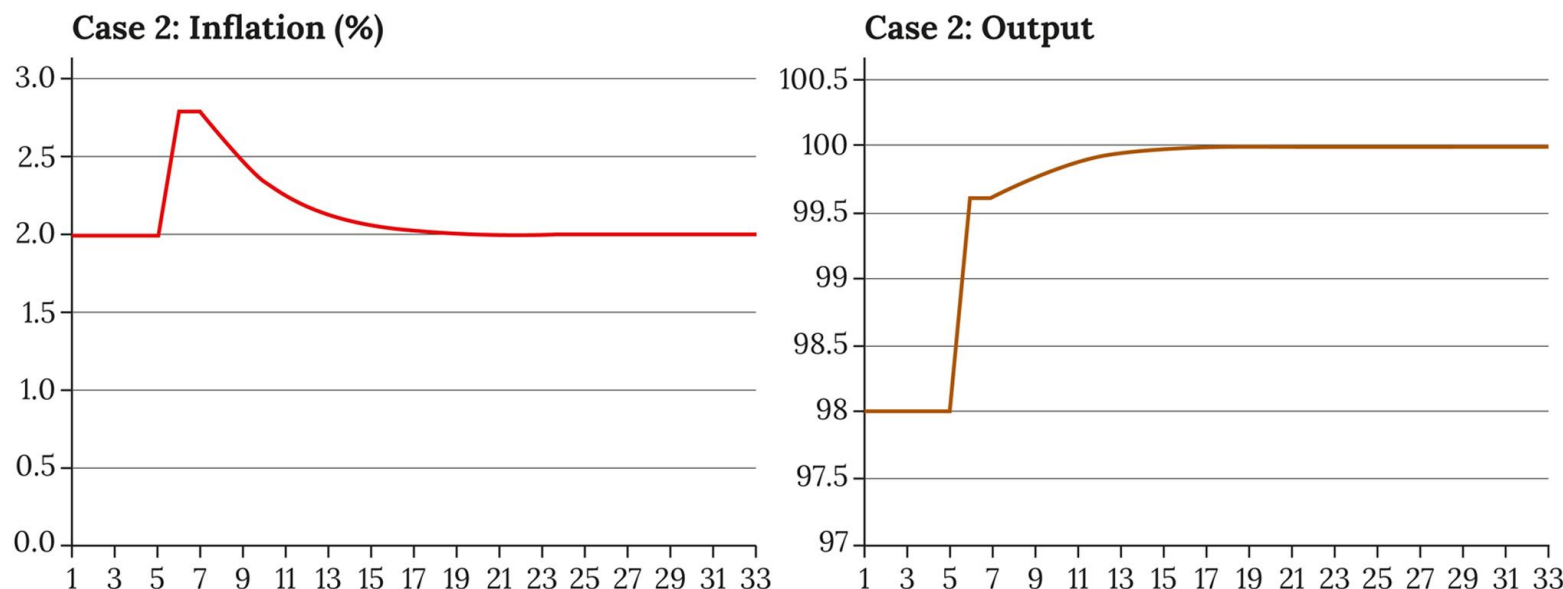


Figure 4.10 Case 2 (the basic hysteresis case). The x-axis represents the number of periods elapsed, starting in period 1. The y axis on the left-hand side graphs represents inflation in the model, in percent increase in prices per period, while the y-axis on the right-hand side graphs represents output per period relative to initial output of 98.

Hysteresis: the Lucas critique

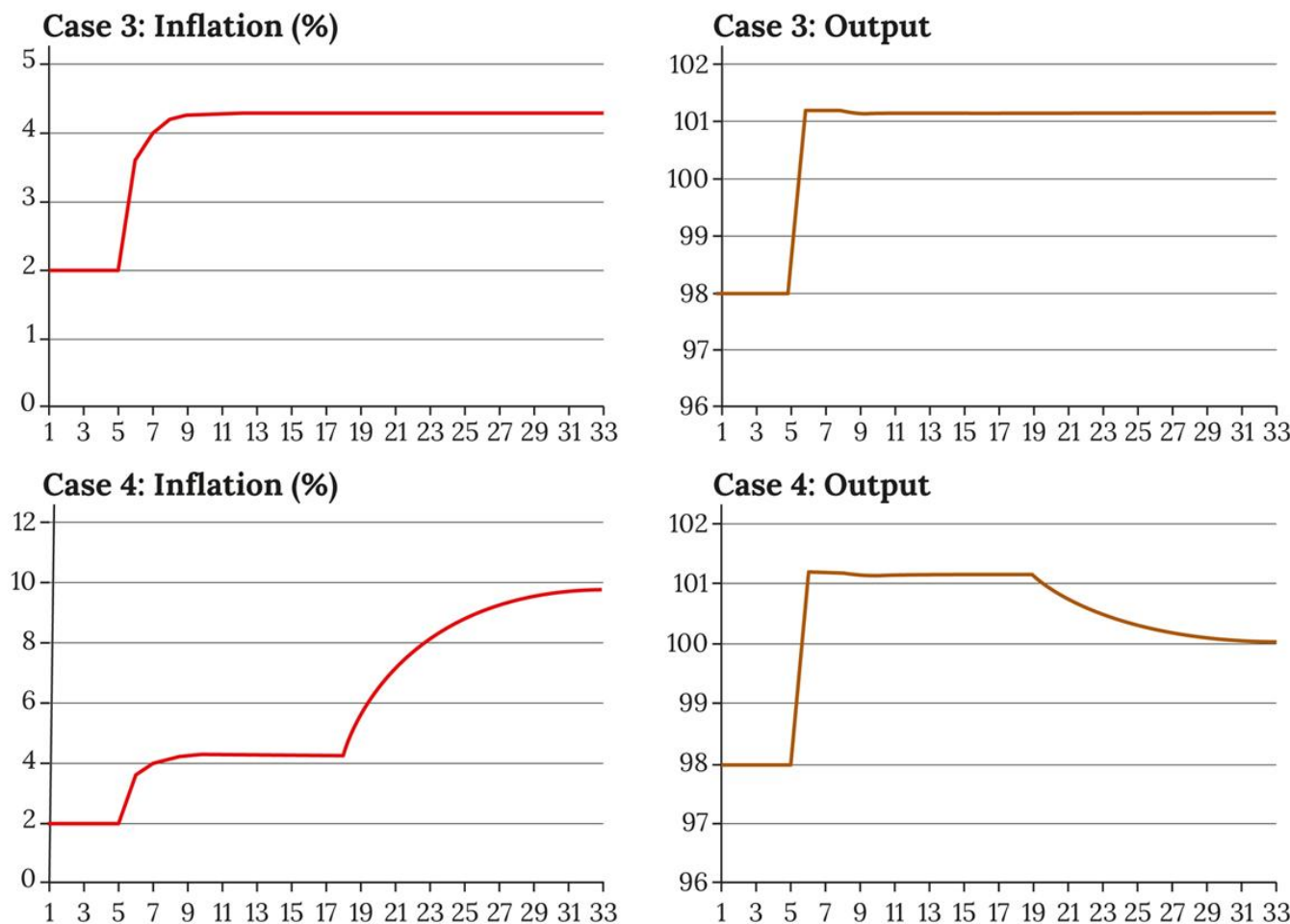


Figure 4.11 Case 3 (supply-side constraints on maximum equilibrium output) and Case 4 (expectations becoming unanchored). The x-axis represents the number of periods elapsed, starting in period 1. The y-axis on the left-hand side graphs represents inflation in the model, in percent increase in prices per period, while the y-axis on the right-hand side graphs represents output per period relative to initial output (98 in Case 3 and 4).

While Case 2's result is striking, the Lucas critique questions its achievability:

- The supply-side imposes a limit (y_e^{max}) on the maximum equilibrium output level attainable and restricts the hysteresis process from operating
 - Case 3: there's an inflation-output tradeoff again
- As the economy moves towards the new equilibrium, the few periods with inflation above target may de-anchor inflation expectations
 - Case 4: agents don't believe in CB's inflation target and χ falls to zero

Hysteresis: a negative demand shock

Through different scenarios under a negative demand shock, we can analyze the contributions and limitations of the model with anchoring and hysteresis:

- Case 5 (up to period 12): without output targeting, in the new equilibrium, output is permanently lower, and inflation is at target
- Case 5 (after period 12): the CB targets the pre-recession equilibrium output level. In the new equilibrium, output is back at pre-recession level and inflation is at target

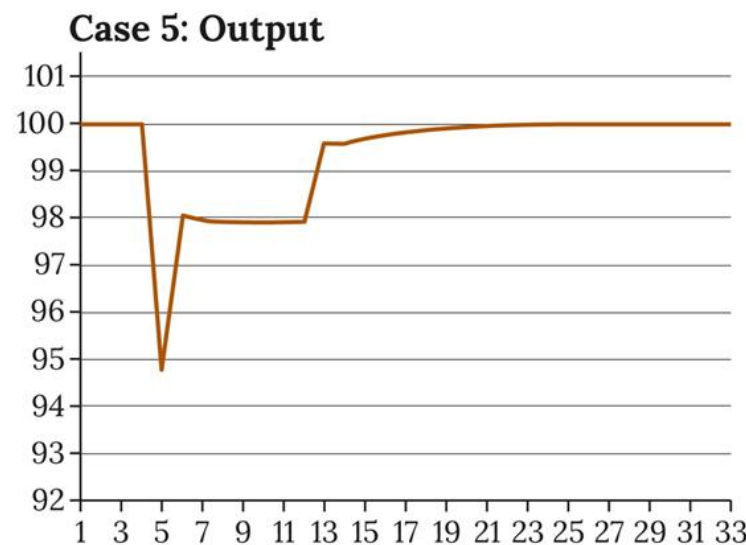
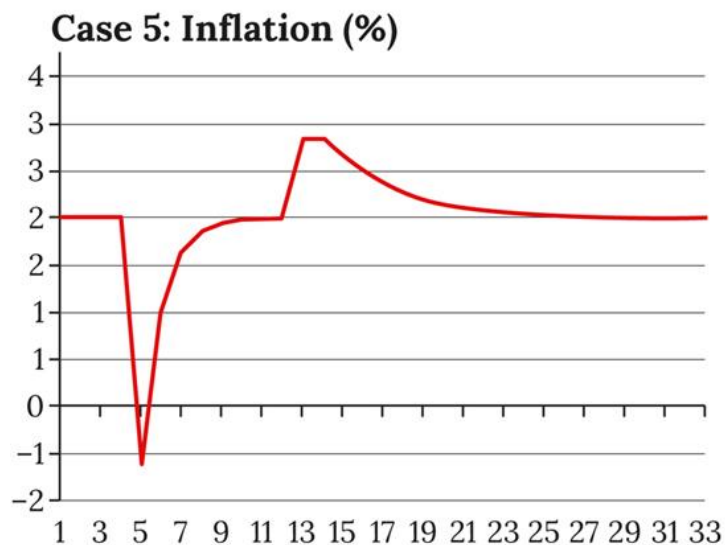


Figure 4.11 Case 5 (responding to a recession). The x-axis represents the number of periods elapsed, starting in period 1. The y axis on the left-hand side graphs represents inflation in the model, in percent increase in prices per period, while the y-axis on the right-hand side graphs represents output per period relative to initial output (100 in Case 5).

Hysteresis: a negative demand shock

- Case 6: the CB sets an over-ambitious $y^T > y_e^{max}$
 - Inflation expectations become unanchored after a period in which inflation is persistently above target
 - In the new equilibrium, output is at equilibrium level ($y = y_e < y_e^{max} < y^T$) and there is inflation bias

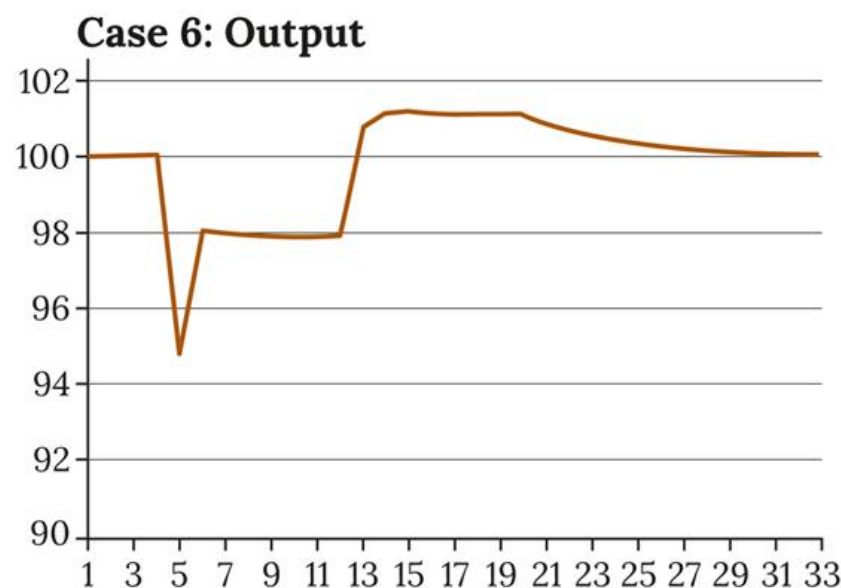
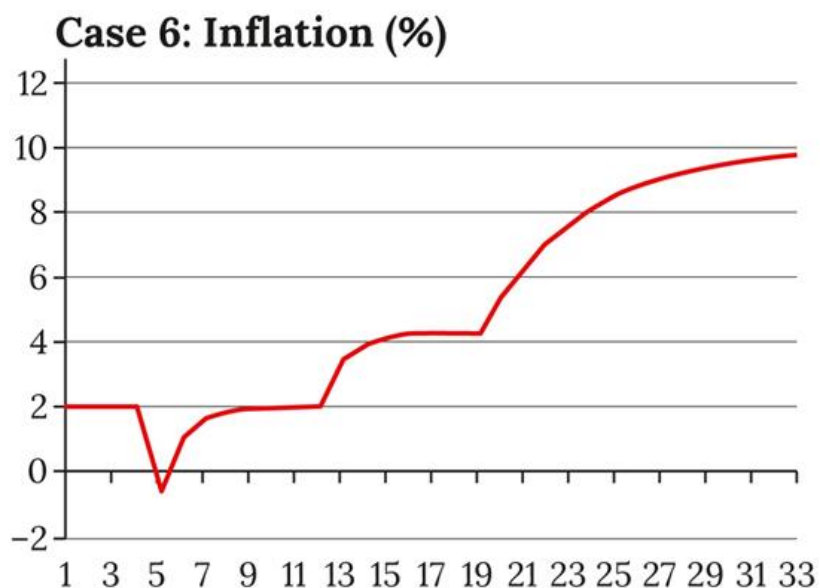


Figure 4.11 Case 6 (an overly ambitious response to a recession). The x-axis represents the number of periods elapsed, starting in period 1. The y axis on the left-hand side graphs represents inflation in the model, in percent increase in prices per period, while the y-axis on the right-hand side graphs represents output per period relative to initial output (100 in Case 6).

Hysteresis: a negative demand shock

The model provides insights on past macroeconomic performance and policies

- Adoption of inflation targeting in 1990s → inflation expectations of agents became anchored at CB's inflation target
- Negative demand shock after the 2008 financial crisis → combination of hysteresis and anchored expectations
 - Without output targeting, economies end with output permanently lower and with inflation at target
 - With pre-recession output level as the CB's target (recognizing the hysteresis process), economies end up with a less costly outcome: output at pre-recession level and inflation at target
- Implications for other deep recessions (e.g., Covid-19 and Ukraine war)
 - Highlights the importance of taking account of hysteresis and the supply-side constraint on any output target

Summary

There are four ways of modelling inflation expectations:

- Static expectations, adaptive expectations, rational expectations and anchored expectations

Under rational expectations, the CB's job is easier as they can influence π^E directly

It is important for CB to communicate their monetary policies to the public:

- CB's transparency and credibility anchors π^E closer to π^T and hence adjustments to shocks are less costly

The Lucas Critique alerts us to the risks of assuming invariant behavior when formulating policy:

- De-anchoring of inflation expectations is possible

Summary

Inflation bias is caused by the CB targeting output above its equilibrium:

- It occurs irrespective of rational or adaptive expectations
- A solution to this is an independent CB

With anchored inflation expectations and hysteresis, CB can achieve output target above equilibrium with inflation at target:

- This highlights the costs of austerity arising from hysteresis
- However, there is a supply-side constraint on output target
- An over-ambitious output target will lead to inflation bias